Exp. No: 11

Verification of Maxwell's Equations and Displacement Current in Time-Varying Electromagnetic Fields

Aim

To verify the Ampère–Maxwell law and the point-form Maxwell's equations for time-varying fields by determining the displacement current in a parallel-plate capacitor, comparing conduction and displacement current densities in different media, numerically checking Faraday's law and the Ampère Maxwell law for a uniform plane wave, and computing the magnetic field produced by the displacement current

Objectives

1. To determine the displacement current density and displacement current in a capacitor driven by a sinusoidal voltage and show that it equals the conduction current in the leads.

2. To compare the conduction-to-displacement current ratio $\sigma/(\omega\varepsilon)$ for a conductor, a lossy dielectric and a poor conductor and to find the crossover frequency.

3. To verify numerically that the electric and magnetic fields of a uniform plane wave satisfy Maxwell's curl equations.

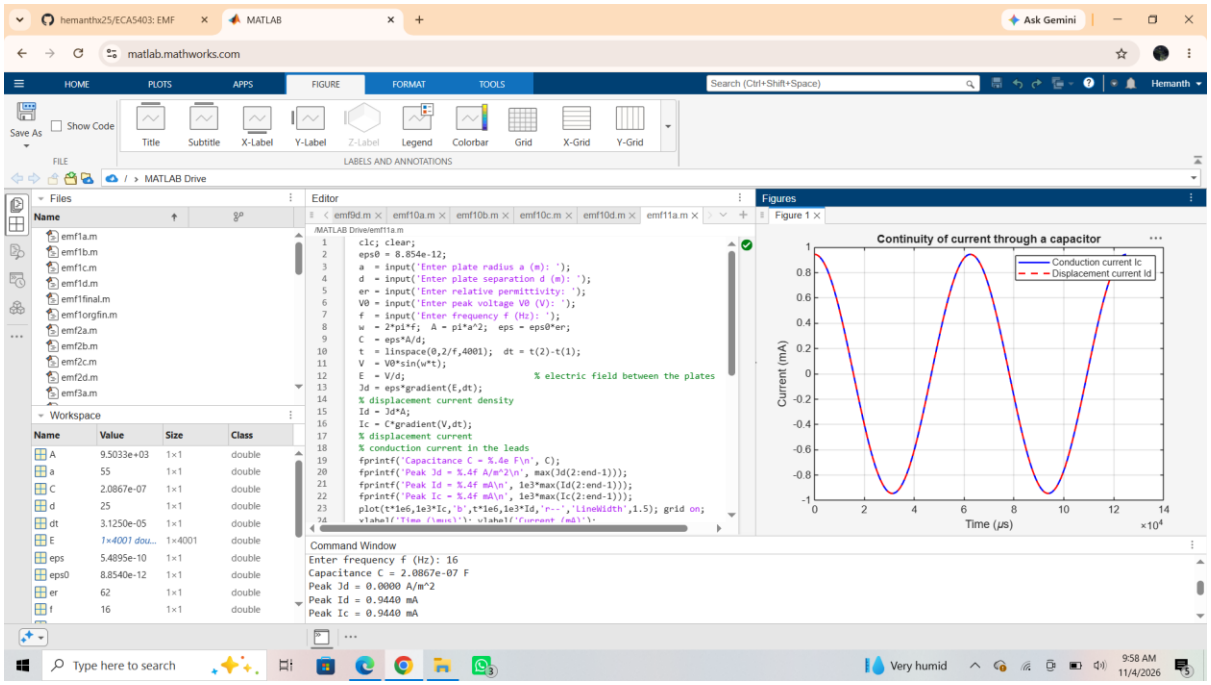
4. To determine the magnetic field produced by the displacement current between and outside the capacitor plates using the Ampère–Maxwell law.

Test Case (i) – Displacement Current in a Parallel-Plate Capacitor

MATLAB Program:

```
clc; clear;
eps0 = 8.854e-12;
a  = input('Enter plate radius a (m): ');
d  = input('Enter plate separation d (m): ');
er = input('Enter relative permittivity: ');
V0 = input('Enter peak voltage V0 (V): ');
f  = input('Enter frequency f (Hz): ');
w  = 2*pi*f;  A = pi*a^2;  eps = eps0*er;
C  = eps*A/d;
t  = linspace(0,2/f,4001);  dt = t(2)-t(1);
V  = V0*sin(w*t);
E  = V/d;                       % electric field between the plates
Jd = eps*gradient(E,dt);        % displacement current density
Id = Jd*A;                      % displacement current
Ic = C*gradient(V,dt);          % conduction current in the leads
fprintf('Capacitance C = %.4e F\n', C);
fprintf('Peak Jd = %.4f A/m^2\n', max(Jd(2:end-1)));
fprintf('Peak Id = %.4f mA\n', 1e3*max(Id(2:end-1)));
fprintf('Peak Ic = %.4f mA\n', 1e3*max(Ic(2:end-1)));
plot(t*1e6,1e3*Ic,'b',t*1e6,1e3*Id,'r--','LineWidth',1.5); grid on;
xlabel('Time (\mus)'); ylabel('Current (mA)');
legend('Conduction current Ic','Displacement current Id');
title('Continuity of current through a capacitor');
```

Output:

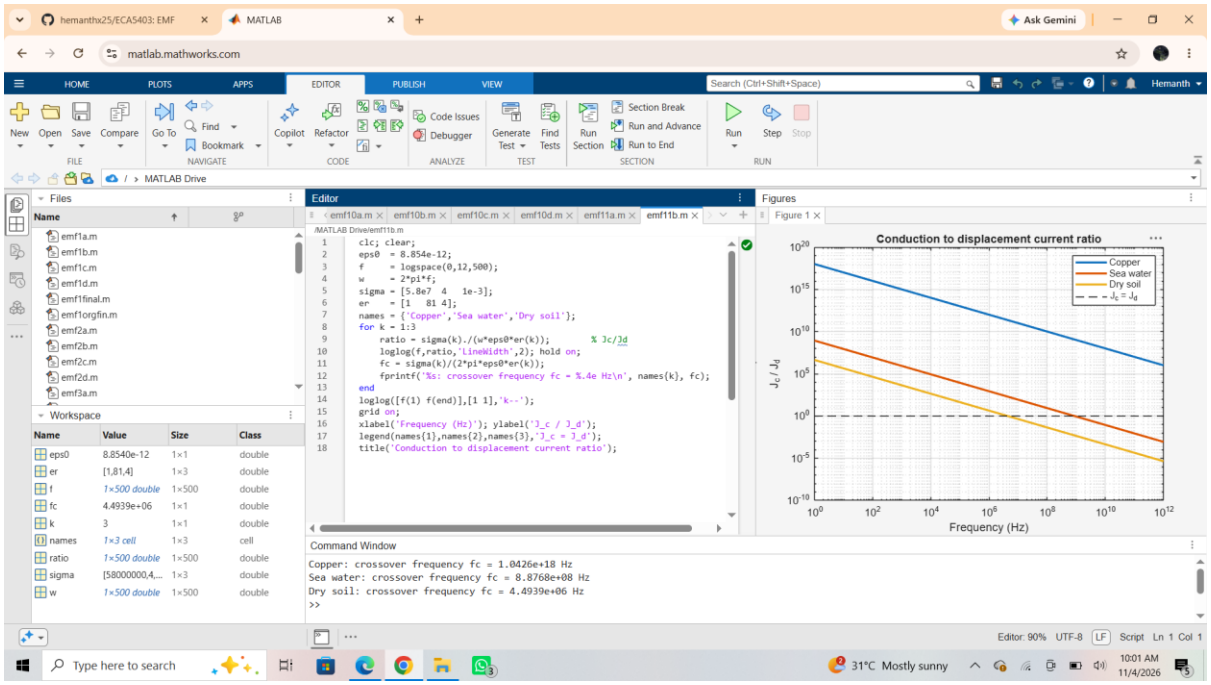


Test Case (ii) – Conduction and Displacement Current in Different Media

MATLAB Program:

```
clc; clear;
eps0  = 8.854e-12;
f     = logspace(0,12,500);
w     = 2*pi*f;
sigma = [5.8e7  4   1e-3];
er    = [1      81  4   ];
names = {'Copper','Sea water','Dry soil'};
for k = 1:3
    ratio = sigma(k)./(w*eps0*er(k));       % Jc/Jd
    loglog(f,ratio,'LineWidth',2); hold on;
    fc = sigma(k)/(2*pi*eps0*er(k));
    fprintf('%s: crossover frequency fc = %.4e Hz\n', names{k}, fc);
end
loglog([f(1) f(end)],[1 1],'k--');
grid on;
xlabel('Frequency (Hz)'); ylabel('J_c / J_d');
legend(names{1},names{2},names{3},'J_c = J_d');
title('Conduction to displacement current ratio');
```

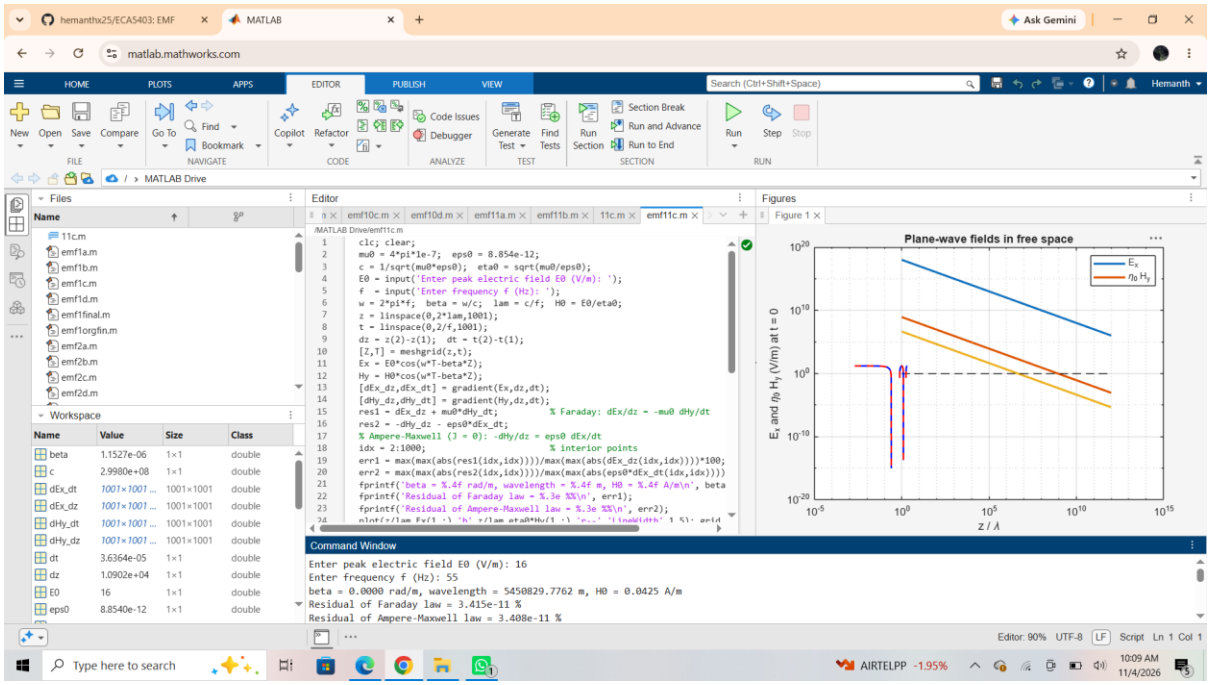
Output:



Test Case (iii) – Numerical Verification of Maxwell's Curl Equations for a Plane Wave

```
clc; clear;
mu0 = 4*pi*1e-7;  eps0 = 8.854e-12;
c = 1/sqrt(mu0*eps0);  eta0 = sqrt(mu0/eps0);
E0 = input('Enter peak electric field E0 (V/m): ');
f  = input('Enter frequency f (Hz): ');
w = 2*pi*f;  beta = w/c;  lam = c/f;  H0 = E0/eta0;
z = linspace(0,2*lam,1001);
t = linspace(0,2/f,1001);
dz = z(2)-z(1);  dt = t(2)-t(1);
[Z,T] = meshgrid(z,t);
Ex = E0*cos(w*T-beta*Z);
Hy = H0*cos(w*T-beta*Z);
[dEx_dz,dEx_dt] = gradient(Ex,dz,dt);
[dHy_dz,dHy_dt] = gradient(Hy,dz,dt);
res1 = dEx_dz + mu0*dHy_dt;          % Faraday: dEx/dz = -mu0 dHy/dt
res2 = -dHy_dz - eps0*dEx_dt;        % Ampere-Maxwell (J = 0): -dHy/dz = eps0 dEx/dt
idx = 2:1000;                        % interior points
err1 = max(max(abs(res1(idx,idx))))/max(max(abs(dEx_dz(idx,idx))))*100;
err2 = max(max(abs(res2(idx,idx))))/max(max(abs(eps0*dEx_dt(idx,idx))))*100;
fprintf('beta = %.4f rad/m, wavelength = %.4f m, H0 = %.4f A/m\n', beta, lam, H0);
fprintf('Residual of Faraday law = %.3e %%\n', err1);
fprintf('Residual of Ampere-Maxwell law = %.3e %%\n', err2);
plot(z/lam,Ex(1,:),'b',z/lam,eta0*Hy(1,:),'r--','LineWidth',1.5); grid on;
xlabel('z / \lambda'); ylabel('E_x and \eta_0 H_y (V/m) at t = 0');
legend('E_x','\eta_0 H_y');
title('Plane-wave fields in free space');
```

Output:

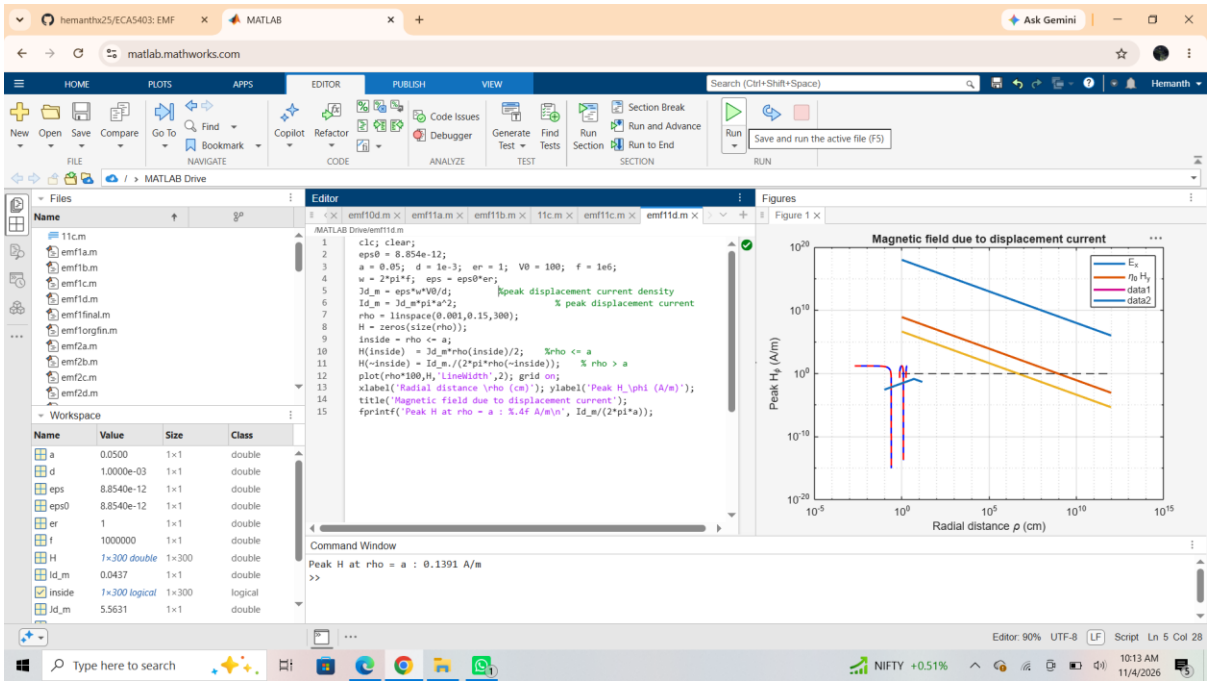


Test Case (iv) – Magnetic Field Produced by Displacement Current (Ampère–Maxwell Law)

MATLAB Program:

```
clc; clear;
eps0 = 8.854e-12;
a = 0.05;  d = 1e-3;  er = 1;  V0 = 100;  f = 1e6;
w = 2*pi*f;  eps = eps0*er;
Jd_m = eps*w*V0/d;                % peak displacement current density
Id_m = Jd_m*pi*a^2;               % peak displacement current
rho = linspace(0.001,0.15,300);
H = zeros(size(rho));
inside = rho <= a;
H(inside)  = Jd_m*rho(inside)/2;          % rho <= a
H(~inside) = Id_m./(2*pi*rho(~inside));   % rho > a
plot(rho*100,H,'LineWidth',2); grid on;
xlabel('Radial distance \rho (cm)'); ylabel('Peak H_\phi (A/m)');
title('Magnetic field due to displacement current');
fprintf('Peak H at rho = a : %.4f A/m\n', Id_m/(2*pi*a));
```

Output:



S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify Maxwell relations and assumptions	Identify Maxwell relations and assumption s	Selects the appropriate integral/differential equation and identifies time varying quantities.	Selecti n is correct with minor omission .	Needs guidance choosing relations.	Uses inappropriate equations.	25	2.5	
2	Calculate displacement current and field quantities	Calculate displacement current and field quantities	Implements the displacement current relation and associated field calculations correctly.	Calculations are mostly correct with minor errors.	Partial calculations require guidance.	Calculations are incorrect or absent.	25	2.5	
3	Verify equations numerically or graphically	Verify equations numerically or graphically	Demonstrates consistency of the chosen Maxwell relation with computed values/plots and units.	Verification is mostly correct.	Verification is incomplete.	No valid verification is demonstrated.	30	3	
4	Explain physical significance	Explain physical significance	Explains displacement current and its role in time varying fields	Explanation is generally correct.	Explanation is superficial or	Cannot explain the result.	20	2	

			with clear documentation.		incomplete.				
Total							100	10	